

Crop Tracking System — Mobile Application & Offline Synchronization Blueprint

This document details the software architecture, technical specifications, and offline-first data synchronization strategy for the **Crop Tracking Mobile Application**.

1. Project Overview & Mobile Strategy

The mobile client provides field workers, farmers, and agronomists with reliable crop monitoring capabilities in areas with limited or intermittent internet connectivity. The core emphasis of this second component is local-first data capture, on-device ML inference for plant disease detection, and robust background synchronization.

2. Offline-First Architecture & Data Sync

To ensure seamless usability in remote farm locations, the application utilizes a local embedded database that queues changes and synchronizes with the server when connectivity is re-established.

Component	Technology & Approach	Purpose / Function
Local Storage	SQLite / Room / WatermelonDB	Store field notes, crop status records, and image references locally.
Sync Queue	Event-driven Change Log Engine	Record local mutations with timestamp tags and optimistic UI updates.
Conflict Resolution	Last-Write-Wins (LWW) / Field Merge	Resolve state discrepancies between local changes and

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		backend API records.
Background Tasks	WorkManager / BackgroundFetch	Perform automatic sync retries when network state becomes online.

3. Mobile Feature Set

- **Field Log Entry:** Capture observations, growth stage updates, photos, and GPS tags directly in the field.
- **Image Classification Model:** Run lightweight TensorFlow Lite / ONNX models on-device for initial offline crop health assessment.
- **Sync Status Dashboard:** Display pending upload queues, last sync timestamp, and network connectivity indicators.
- **Task Notifications:** Local scheduled push notifications for upcoming irrigation, application, or harvest tasks.

4. Mobile Data Payload Example

```
{
  "sync_id": "sync_9876543210",
  "device_id": "device_android_01",
  "timestamp": "2026-10-08T00:19:00Z",
  "mutations": [
    {
      "action": "CREATE",
      "entity": "CropObservation",
      "data": {
        "observation_id": "obs_12345",
        "crop_id": "crop_88",
        "leaf_health_score": 0.85,
        "disease_detected": "Early Blight",
        "confidence": 0.92,
```

```
"photo_local_uri": "file:///storage/emulated/0/Pictures/crop_1.jpg",  
"synced": false  
}  
}  
]  
}
```

5. Mobile Development Milestones

1. **Phase 1: Local Database Setup & UI Wireframing** — Local SQLite schema definition, mobile UI layout setup.
2. **Phase 2: Offline Capture & Sync Queue Implementation** — Queueing system, network listener, retry logic.
3. **Phase 3: Image Classification Integration** — Deploying lightweight ML model for offline leaf diagnostics.
4. **Phase 4: End-to-End Field Testing & Sync Validation** — Testing sync conflict scenarios and offline stability.